

Lab 4 Report: Programming Symmetric & Asymmetric Crypto

Features Implemented

1. AES Encryption/Decryption with 128-bit and 256-bit keys in ECB and CFB modes
2. RSA Encryption and Decryption using OAEP padding
3. RSA Digital Signatures with PKCS#1 v1.5 padding
4. SHA-256 Hashing for file integrity verification
5. Performance Analysis with graphical visualization
6. Key Management with persistent storage

Implementation Details

1. Key Generation and Management

The program generates and manages three types of keys:

- AES-128 key: 16 bytes (128 bits)
- AES-256 key**: 32 bytes (256 bits)
- RSA key pair: 2048 bits (default)

2. AES Implementation

Modes Supported:

- ECB (Electronic Codebook): Simple block cipher mode
- CFB (Cipher Feedback): Stream cipher mode with IV

Key Features:

- Automatic padding for ECB mode
- Random IV generation for CFB mode
- Encrypted data stored in binary files
- Support for both 128-bit and 256-bit keys

3. RSA Implementation

Operations:

- Encryption/Decryption: Using OAEP padding for security
- Digital Signatures: PKCS#1 v1.5 signing scheme
- Key Management: Separate public/private key files

4. SHA-256 Hashing

- Computes cryptographic hash of files
- Outputs hexadecimal digest
- Useful for file integrity verification

5. Performance Analysis

The program measures and analyzes:

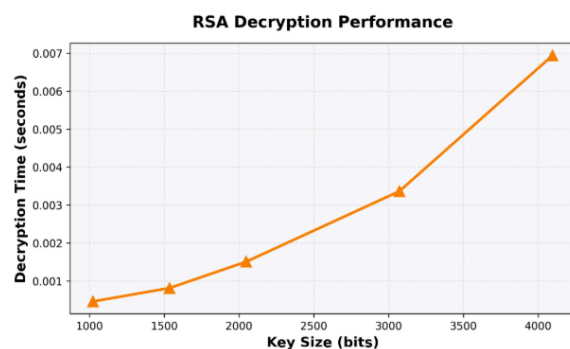
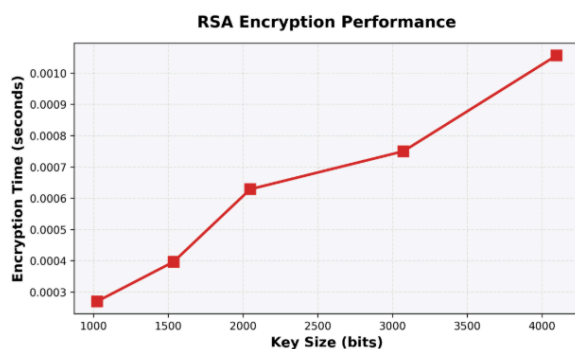
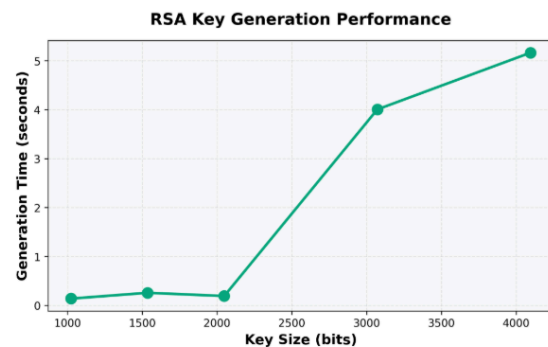
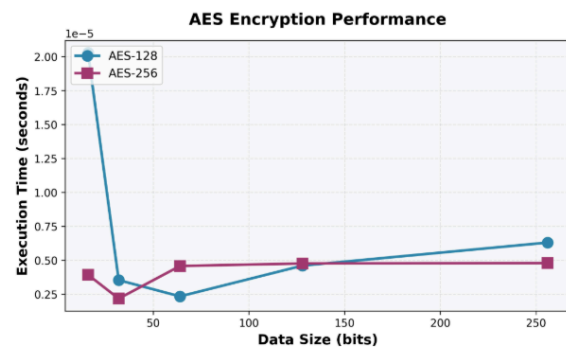
- AES encryption time vs. data size

- RSA key generation time vs. key size
- RSA encryption time vs. key size
- Generates comparative graphs

Prerequisites

`pip install pycryptodome matplotlib`

Performance Analysis Results



Resources and References

Code Resources:

1. PyCryptodome Documentation: <https://pycryptodome.readthedocs.io/>
2. AES Implementation Guide: <https://www.pycryptodome.org/en/latest/src/cipher/aes.html>
3. RSA Implementation Guide: https://www.pycryptodome.org/en/latest/src/public_key/rsa.html
4. LLM-Claude, Deepseek

Educational Resources:

1. Python Cryptography Tutorials: <https://www.laurentluce.com/posts/python-and-cryptography-with-pyrypto/>
2. Cryptography Concepts: Course materials and openssl documentation